Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University College Dublin
Short name / acronym used in projects	UCD
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	https://www.ucd.ie/conway/
Country	Ireland
Organisation type	x Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	UCD Belfield, Dublin D04 V1W8, Ireland
Department(s), institute(s), research group(s) or unit(s) that would be involved	UCD School of Biomolecular & Biomedical Science UCD Conway Institute
Additional institutional resources relevant to collaboration	UCD Conway Genomics Core (https://www.ucd.ie/conway/technology/genomicscore/) UCD Conway Metabolomics Core (https://www.ucd.ie/conway/technology/metabolomics/) UCD Aquatics Facility

Main contact person for this profile	Derek Costello, Associate Professor, Neuropharmacology, UCD
Main contact email	Derek.costello@ucd.ie

Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
x Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
x Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	☐ Ocular vascular disease
x Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	x Other (please specify): Neuroinflammation, neurodegeneration and neurotoxicity

Free Keywords for internal searchable database (please provide up to 10)	Microglial culture, neuronal culture, zebrafish, metal dyshomeostasis, novel pharmacotherapeutics, microglial dysfunction, neurodegeneration models, Alzheimer's disease, Parkinson's disease, temporal lobe epilepsy, bacterial meningitis
Geographies of interest for exchanges (optional)	Spain; Sweden

Overview of your research / innovation focus
Our research is focused on understanding the cellular mechanisms underlying neuroinflammatory, neurodevelopmental and neurodegenerative disease, along with their primary risk factors. In particular, we investigate the immune dysregulation that drives the chronic neuroinflammation characteristic of neurodegenerative disease pathology. We use cultured mammalian and human microglia and macrophages to evaluate the inflammatory changes, metabolic dysfunction and reprogramming indicative of the dysfunctional microglia in the diseased brain. Using human and mammalian neuronal cultures we further explore the mechanisms underlying the neuronal injury and neurotoxicity characteristic of the neurodegenerating brain. We are particularly interested in the role of metal ion imbalance in promoting neuronal death. Accumulation of ions such as iron and copper in the brain is a hallmark of neurodegenerative disease, driving neuronal dyshomeostasis and cell death mechanisms including ferroptosis and cuproptosis. In parallel, we use model organisms, including zebrafish and c. elegans, to generate in vivo models of neuroinflammatory and neurodegenerative disease, including Alzheimer's disease, Parkinson's disease and epilepsy. We directly explore the toxicity and behavioural impairment that typify disease states, and further validate our in vitro analyses. Importantly, we use these in vitro and in vivo models to explore novel therapeutic targets and pharmacotherapeutic strategies for neurodegenerative disease. Currently there are limited/no effective treatments for neurodegenerative disease. Most attempts to date have focused on a single aspect of the pathology, and have resulted in minimal or no clinical efficacy. Our research explores strategies that target multiple facets of pathology, such as inflammation, oxidative injury, metal overload, metabolic imbalance. These strategies include naturally-occurring compounds and novel derivatives, specific inhibitors, Schiff base ligands, and novel small molecule agents.

Why your team could be relevant to a translational eye-health and vision consortium
Our team brings a neurobiological and neuropharmacological perspective to eye health, with particular expertise in mechanisms shared between neurodegenerative disease and retinal/visual-system dysfunction. We provide an experimental pipeline spanning mechanistic cellular studies, multi-omics discovery and therapeutic screening, through to functional in vivo validation. **Mechanistic neurobiology expertise:** We bring broad expertise spanning neuronal and glial physiology, neuroinflammation, neurodegeneration, cellular metabolism and pharmacological intervention, allowing us to interrogate mechanisms contributing to retinal and visual-system dysfunction. **Associating eye disease with neurodegeneration:** Our research addresses pathological processes relevant across retinal and CNS disorders, including neuroinflammation, neuronal dysfunction, metabolic stress and dysregulated cellular homeostasis. **Therapeutic investigation:** We investigate complex disease mechanisms with the aim of identifying actionable therapeutic targets and interventions, providing a bridge between fundamental disease biology and translational development. **Complementary to eye health:** Our experimental neurobiology capabilities can help translate clinical observations and disease-associated signatures into cellular and in vivo models for mechanistic investigation and therapeutic testing. **Therapeutic innovation**: Our experience in pharmacological intervention and development of novel small-molecule approaches provides opportunities to identify and validate multi-target therapeutic strategies appropriate for complex and multifactorial visual disorders.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
Cellular and molecular neurobiology: In vitro models for investigating neuronal and glial function, neuroinflammation, neurotoxicity, metabolic dysfunction and mechanisms of neurodegeneration. **In vivo disease models and functional phenotypes**: Established models for translation of molecular and cellular findings to functional outcomes, including in vivo toxicity, behavioural phenotypes and visual assessment. **Multi-omic approaches:** Proteomic, transcriptomic and metabolomic approaches for pathway analysis and identification of disease biomarkers, molecular signatures and novel therapeutic targets. **Molecular and biochemical profiling:** Gene and protein expression analysis, cellular metabolic assessment, inflammatory signalling, oxidative-stress and cell-death pathways, and investigation of cellular homeostatic mechanisms. **Pharmacological screening and validation:** Experimental platforms for screening and characterising candidate therapeutic compounds across in vitro and in vivo models. **Small-molecule therapeutic development:** Expertise and access to novel and patent-protected compounds, including multi-target pharmacological approaches with potential for therapeutic development and repurposing. **Integrated translational pipeline:** Capacity to progress findings from pathway identification and cellular validation to pharmacological intervention and functional validation. This provides experimental support for clinically derived hypotheses and candidate therapeutics.

Infrastructure, equipment, facilities or datasets relevant to consortium work
Our team are based at the UCD Conway Institute of Biomolecular & Biomedical Research. (https://www.ucd.ie/conway) The UCD Conway Institute houses a state-of-the-art aquatics facility for zebrafish experimentation. This includes equipment for behavoiural assessment such as measurement of visual motor function and optokinetic response, microinjection equipment for embryonic and adult pharmacological interventions, and microscopy/imaging analysis. Conway core facilities include a genomics facility for gene expression and transcriptomic analysis, a proteomics core facility, metabolomics core for analysis of metabolic signatures, core technologies including *Seahorse* analysers and *Incucyte* imagers for live cell analysis of cell health, metabolism and respiration, and a dedicated electrophysiology suite for analysis of neuronal function in ex vivo brain tissue.

Newly-upgraded tissue culture suites allow dedicated facilities for culture and maintenance of cell lines, primary culture systems and human iPSC-derived cells and organoids.

Training capacity and knowledge-transfer potential

Skills and training opportunities for inbound staff:

Interdisciplinary training environment: Inbound staff will join the highly interdisciplinary and collaborative environment of the UCD Conway Institute, gaining exposure to expertise spanning neurobiology, neuroinflammation, neurodegeneration, visual dysfunction, pharmacology and therapeutic development.

Preclinical disease modelling: Hands-on training in the development, maintenance and application of in vitro neuronal and glial models and in vivo zebrafish models of neurological and visual disorders, including Alzheimer's and Parkinson's diseases, temporal lobe epilepsy, bacterial meningitis and inherited blindness.

Cellular and molecular analysis: Training in experimental approaches for investigating neuronal and glial toxicity, neuroinflammation, cellular metabolism, metal dyshomeostasis and disease-associated molecular signalling, together with interpretation and integration of molecular datasets.

Functional phenotyping: Practical experience linking molecular and cellular pathology with in vivo functional outcomes, including behavioural, neurological and visual/functional assessment in zebrafish disease models.

Experimental design and translational validation: Training in the design of complementary in vitro and in vivo studies, selection of appropriate experimental endpoints, and translation of mechanistic findings across model systems.

Therapeutic screening and pharmacological evaluation: Experience in screening and mechanistic characterisation of candidate therapeutics, including assessment of efficacy, toxicity and modulation of disease-associated pathways.

Novel therapeutic development: Opportunities to investigate multifunctional small molecules developed with pharmaceutical chemistry collaborators, alongside naturally occurring bioactive compounds, providing experience in progressing candidate interventions from mechanistic rationale through preclinical evaluation.

Translational research skills: Exposure to the wider pathway from disease-mechanism discovery and target identification through therapeutic validation, interdisciplinary collaboration and development of research findings with potential for therapeutic translation.

Transferable research skills: Development of competencies in experimental planning, critical data interpretation, scientific communication and collaborative working across disciplines. Our team routinely engages with scientific outreach and educational activities, along with patient/advocate representatives.

What your team would hope to gain from participating

What our team would hope to gain from participating:

Establish collaborations with consortium partners that offer advanced experimental models, in particular imaging technologies, computational approaches to complement our existing neurobiological and pharmacological capabilities.

Access to clinically derived samples and datasets, to further validate mechanisms identified in our experimental models and establish their relevance to human disease.

Opportunities to integrate findings from our cellular and in vivo models with patient-derived cells, iPSC-based systems, organoids etc, to strengthen the translational relevance of our mechanistic and therapeutic findings.

Access to specialist expertise in visual function, retinal biology, imaging and electrophysiology would enable us to extend our existing neurobiological models and develop more sophisticated approaches for evaluating visual-system dysfunction and therapeutic rescue.

We would particularly value partnerships that can help progress promising small molecules and other interventions beyond early preclinical discovery, including medicinal chemistry, pharmacokinetics, drug-delivery, toxicology and later-stage preclinical validation.

We hope to identify mechanisms shared between visual disorders and wider neurological, inflammatory, metabolic and age-related diseases, creating opportunities for therapeutic repurposing and development of interventions with relevance beyond ophthalmology.

Participation would provide a foundation for exchange expertise, joint publications and development of collaborative and cross-disciplinary funding applications.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Derek Costello	Associate Professor and Principal Investigator	UCD School of Biomolecular & Biomedical Science (SBBS)	Researcher	Experienced	Derek.costello@ucd.ie
Jodie Kearney	PhD Student	UCD SBBS	Researcher	Early-stage	Jodie.kearney@ucdconnect.ie
TBD	2x doctoral students	UCD SBBS	Researcher	Early-stage	

Main disciplines represented in your team	Neurobiology, pharmacology, immunology, neurodegeneration, neurotherapeutics
Experience in EU / international / multi-partner collaborative projects	Participation as a research partner in the *Crystal3* consortium. I supervised a 12-month secondment at UCD Conway, and a 3-month staff exchange from our team. To date this has yielded 2 publications, and 2 currently under review.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes x No ☐ Not applicable | Short note: Primarily preclinical research, however routine engagement with PPI/patient advocates to inform research.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Clarke NC, McCabe E, Jensen LD, Creaven BS, Costello DA (2025) Salicylaldehyde benzoylhydrazone protects against ferroptosis in models of neurotoxicity and behavioural dysfunction, in vitro and in vivo. J Mol Neurosci 75:2. doi: 10.1007/s12031-025-02371-2

Galvin DM, Fernandez-Garcia S, Dawson E, Pryce C, Egan BP, Clarke NC, Reynolds AL, Costello DA (2025) Ergolide regulates microglial activation and inflammatory-mediated dysfunction: a role for the cysteinyl leukotriene pathway. Int Journal Mol Sci 26:11. doi: 10.3390/ijms26115050

Al-Thani NA, Stewart GS, Costello DA (2025) The role of the urea cycle in the Alzheimer's disease brain. J Neurochem 169:3. doi: 10.1111/jnc.70033

Al-Thani NA, Zinck D, Stewart GS, Costello DA (2024) Modulation of urea transport attenuates TLR2-mediated microglial activation and upregulates microglial metabolism in vitro. Metabolites 14:11. doi: 10.3390/metabo14110634

Howe AM, Burke S, O'Reilly ME, McGillicuddy FC, Costello DA (2022) Palmitic acid and oleic acid differently modulate TLR2-mediated inflammatory responses in microglia and macrophages. Molecular Neurobiology 59:2348–2362. doi: 10.1007/s12035-022-02756-z

Howe AM, Cosgrave A, Ó'Murchú M, Britchfield C, Mulvagh A, Fernandez-Perez I, Dykstra M, Jones AC, Costello DA (2020) Characterising lipoteichoic acid as an in vitro model of acute neuroinflammation. International Immunopharmacology 85:106619. doi: 10.1016/j.intimp.2020.106619

Up to 5 relevant previous projects, grants, networks or collaborations

Title: Exploring coumarin-derivative Shift base compounds as potential therapeutics for bacterial meningitis
Funder:Research Ireland (IRC)
Funder reference: GOIPG/2025/8836
01 Sep 2025 - 31 Aug 2029
Title: Investigation of novel neurotherapeutic strategies for drug-resistant epilepsy

Funder:Irish Research Council (IRC)
Funder reference: GOIPG/2024/3286
01 Sep 2024 - 31 Aug 2027
Title: Explore the therapeutic potential of seaweed extracts for Alzheimer's disease-related neurodegeneration
Funder: Institutional
01 Sep 2023 - 31 Oct 2025
Title: Commercial & Research Opportunity for Cysteinyl Leukotriene Signalling in Ocular & CNS Dysfunction Cancer and Cardiovascular Disease (CRYSTAL3)
Funder reference:101007931
Funder: Horizon 2020
01 Jan 2021 - 31 Dec 2024
Title: Investigating the therapeutic potential of phytocannabinoids in models of neuroinflammation
Funder reference: EPSPG/2020/73
Funder: Irish Research Council (IRC)
01 Apr 2020 - 31 Mar 2022
Title: Investigating the role of UT-B in the pathogenesis of Alzheimer's disease
Funder: Jack Pickard Research Trust (UK)
01 Mar 2019 - 28 Feb 2021

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Title: Novel coumarin-based neuroprotective agents Inventors: Creaven B.S., Costello D.A. Patent number: Application in preparation (Sept 2026)

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	N/A.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Neurobiology and disease models: Integrated in vitro neuronal/glial and in vivo zebrafish models for investigating neurodegeneration, neuroinflammation and visual dysfunction, with molecular, behavioural and functional phenotyping. **Therapeutic discovery and preclinical validation:** Pharmacological screening and mechanistic evaluation of novel, repurposed and naturally derived therapeutic candidates, including access to patent-protected multifunctional small molecules and platforms for in vitro and in vivo efficacy testing. **Mechanistic and multi-omics discovery:** Integrated molecular, transcriptomic, proteomic and metabolomic approaches for identifying disease mechanisms, biomarkers and therapeutic pathways, with particular expertise in neuroinflammation, cellular metabolism and homeostatic dysfunction.

Capabilities or expertise you would especially like to access through the consortium

Clinical datasets and patient-derived resources: Access to clinical datasets and disease-associated biomarker data to validate mechanisms identified in our experimental models and establish their relevance to human disease. **Computational biology:** Expertise in bioinformatics, systems biology and advanced computational approaches for integration and interrogation of transcriptomic, proteomic, metabolomic and clinical datasets. **Medicinal and pharmaceutical chemistry**: Expertise in medicinal chemistry, structure–activity relationships and lead optimisation to support progression of promising therapeutic compounds from biological proof-of-concept towards candidates with drug-like properties. **Drug development and delivery:** Complementary expertise in pharmacokinetics, drug delivery etc to strengthen the translational pathway for novel and repurposed therapeutic candidates. **Human-relevant models and phenotypes:** Access to patient-derived cells, iPSC-based systems etc, and assessment of visual and neurological function to bridge our existing cellular and in vivo platforms with human disease.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
N/A

9. Additional collaborators you would like us to consider

If there are any additional organisations, teams or contacts - current collaborators, previous collaborators or desired future collaborators - that you think could be relevant for this MSCA-SE consortium, please list them below. The coordination team may contact them to explore their potential interest in joining the consortium-building process.

Please provide the following details (as many entries as needed).

Type of organisation: ☐ Academic ☐ Clinical ☐ SME/industry ☐ NGO/foundation ☐ Public body ☐ Other

Organisation / team name:

Country:

Website:

Existing link with your team: ☐ Current collaborator ☐ Previous collaborator ☐ Desired collaborator ☐ Other

Suggested contact person/email (if known):

Why could they be relevant for this MSCA-SE consortium?

Any additional comments or sensitivities before contacting them?

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants.

The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

Name: Derek Costello
Date completed: 31/08/26

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